

TOPIC 3: INCOMPLETE DOMINANCE, CO-DOMINANCE & MULTIPLE ALLELES

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: _____

Roll / Reg. No: _____

Date / Batch: _____

SECTION A: Multiple Choice Questions (1 Mark Each)

1. In snapdragon (**Antirrhinum majus**), crossing true-breeding red (RR) with white (rr) yields pink (Rr) F₁ due to:
(A) Co-dominance (B) Incomplete dominance
(C) Pleiotropy (D) Polygenic inheritance
2. ABO blood grouping in humans is an example of:
(A) Co-dominance and Multiple allelism (B) Incomplete dominance
(C) Polygenic inheritance (D) Sex-linked inheritance
3. How many genotypes and phenotypes are possible in human ABO blood group system?
(A) 4 genotypes, 6 phenotypes (B) 6 genotypes, 4 phenotypes
(C) 3 genotypes, 4 phenotypes (D) 6 genotypes, 6 phenotypes
4. In Andalusian fowl, crossing pure black and pure white varieties results in blue-feathered F₁ due to:
(A) Complete dominance (B) Incomplete dominance
(C) Epistasis (D) Multiple alleles

SECTION B: Fill in the Blanks (1 Mark Each)

5. The ABO blood grouping is controlled by the gene designated as _____.
6. A person with blood group AB possesses alleles _____ and shows co-dominance.
7. In cattle, crossing homozygous red (RR) with white (WW) produces _____ coat colour in F₁.
8. In snapdragon, the phenotypic and genotypic ratios in F₂ generation are both identical to _____.

SECTION C: Very Short Answer Questions (2 Marks Each)

9. [2M] Differentiate between Incomplete Dominance and Co-dominance with one example each.

10. [2M] Explain why allele 'i' does not produce surface antigen sugar in ABO blood grouping.

SECTION D: Short Answer Questions (3 Marks Each)

11. [3M] A man with blood group A marries a woman with blood group B. They have a child with blood group O. Work out the genotypes of parents and possible blood groups of future children.

12. [3M] Tabulate all 6 possible genotypes and their corresponding 4 phenotypes in the human ABO blood group system.

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. [5M] Explain Incomplete Dominance with a schematic cross in *Mirabilis jalapa* or *Antirrhinum majus*. Differentiate between dominance and incomplete dominance.

14. [5M] Explain Co-dominance with reference to (a) ABO blood grouping in humans and (b) Coat colour in cattle with schematic crosses.